

Data Handling: Import, Cleaning and Visualisation

Exercise 5: Applied Data Analysis with R

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Exercises¹

Download and unzip `data.zip` (posted along these instructions on StudyNet). The zip-file contains four csv-files with data needed to solve the following exercises. Make sure you store them in the respective working directory of your R session when solving the exercises.

Exercise A

Load `tidyverse` and read `stocks.csv` into R (store it in variable `stocks`). The dataset is currently in long format. Use `spread()` to transform it into wide format (separate columns for different years). Store the resulting `tibble/data.frame` in a variable called `stocks2`. Then, run the following code.

```
gather(stocks2, "year", "return", `2015`:`2016`) == stocks
```

Why is the result not identical to the original dataset (`stocks`)? In other words, Why are `gather()` and `spread()` not perfectly symmetrical? (Hint: look at the variable types and think about column names.)

Exercise B

Load `tidyverse` and read `countries.csv` into R (store it in variable `countries`). Then, run the following code.

```
gather(countries, 1999, 2000, key = "year", value = "cases")
```

Why does this code fail?

Exercise C

Load `tidyverse` and read `people.csv` into R (store it in variable `people`). Try to spread `people`. Why does the spreading fail? How could you add a new column to fix the problem?

```
# sample solution
people <- read_csv("people.csv")
# assuming there are two Philip Woods
people$id <- c(1,1,2,3,3)
spread(people, key, value)
```

Exercise D

Load `tidyverse` and read `preg.csv` into R (store it in variable `preg`). Tidy `preg`. Do you need to spread or gather it? What are the variables?

¹Adopted from <https://r4ds.had.co.nz/tidy-data.html>.

Mock Exam Questions

Part 1: TRUE/FALSE Questions

For each of the following statements, decide whether the statement is TRUE (T) or FALSE (F).

1. The following line of R code will return "bnn".

```
str_split("banana", pattern= "a")
```

Part 2: Multiple Choice Type A

For each of the following questions/statements, decide which of the possible responses are correct. NOTE: Either ONE OR SEVERAL of the response statements can be correct. In order to get a point, all correct response statements need to be marked!

1. Regarding the data semantics (vocabulary used to describe data) introduced in this course, which of the following statements is correct?

- A) An observation contains all values measured on the same unit (e.g., a person).
 - B) A variable contains all values that measure the same underlying attribute across units/observations.
 - C) Every observation belongs to a value and a variable.
 - D) Datasets typically contain more variables than values.
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Part 3: Multiple Choice Type B (only one correct!)

For each of the following questions/statements, decide which of the possible responses is correct. NOTE: For each question only ONE of the possible responses is correct. Select ONLY ONE of the response statements per question!

1. Which of the following statements about ,tidy‘ data is correct?

- A) When a csv-file is imported with `read_csv()`, the function issues a warning if the data imported is not tidy.
- B) Datasets in wide format are generally not considered ,tidy‘.
- C) Data stored in csv-format is generally referred to as ,tidy data‘.
- D) `tibbles` can only contain tidy data.